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RouterInterp: Understanding Superposed Specialisation in Mixture of Experts Routing

Experts Do Not Own Domains—They Activate on Disjoint Unions of Fine-Grained Features, and Sparse Autoencoder Probes Explain Why

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MoE routing does not follow the single-domain assumption: each expert activates on a *disjoint union* of fine-grained features—the Superposed Specialisation Hypothesis. RouterInterp identifies these features via Sparse Autoencoder probes and composes them into natural-language routing explanations, achieving 65% higher routing-detection accuracy than token-statistics baselines on gpt-oss-20b.

AT A GLANCE

Problem: It is unknown why specific tokens are routed to specific experts in MoE language models.

Why it matters: Understanding routing is essential for auditing, debugging, and improving MoE architectures.

Method: Sparse linear probes over SAE features predict routing; top-feature descriptions are composed into expert explanations.

Result: ≈65% higher routing-detection accuracy on gpt-oss-20b vs. token-statistics baselines.

Evidence: Causal interventions confirm the SAE features causally influence routing; manual validation confirms explanation quality.

The Big Picture

Mixture-of-Experts (MoE) models achieve state-of-the-art scaling efficiency by routing each token through a small subset of expert feed-forward networks. Despite widespread deployment, the routing mechanism remains a black box: knowing *why* a token is routed to a particular expert is important for diagnosing failure modes, ensuring balanced expert use, and building trust in deployed systems.

The dominant assumption—"the code expert," "the French expert," etc.—is a convenience fiction. RouterInterp provides empirical and causal evidence for a richer picture: the **Superposed Specialisation Hypothesis (SSH)**.

Why Existing Approaches Fall Short

Token-statistics methods compute the frequency distribution of token types per expert and label each with the most common category. These methods are coarse, conflate co-occurrences with specialisation, and show low prediction accuracy when routing is polysemous.

Probing classifiers on raw activations predict routing but yield no human-readable explanation.

Manual inspection is intractable across the hundreds of experts and billions of parameters in production MoE models.

Core Mechanism

Superposed Specialisation Hypothesis: Each expert e activates not on one broad domain, but on a disjoint union of fine-grained linguistic or semantic features—mirroring the superposition phenomenon known to occur in individual neurons.

RouterInterp operationalises this via Sparse Autoencoders (SAEs):

1. Derive monosemantic SAE features from the residual stream.
2. Fit L_1-penalised probes to predict routing from SAE activations.
3. Compose the top-predictive features' descriptions into a unified expert explanation.

Technical Formulation

Let $\mathbf{h}_t \in \mathbb{R}^d$ be the residual stream at token t. A Sparse Autoencoder maps it to a sparse feature vector:

$$\mathbf{f}_t = \mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}\,\mathbf{h}_t + \mathbf{b}_{\mathrm{enc}}), \quad \mathbf{f}_t \in \mathbb{R}^F$$

The routing indicator for expert e is:

$$r_{t,e} = \mathbf{1}[\text{expert } e \text{ selected for token } t]$$

RouterInterp solves the L_1-penalised logistic regression:

$$\hat{r}_{t,e} = \sigma(\boldsymbol{\beta}_e^\top \mathbf{f}_t), \quad \boldsymbol{\beta}_e = \arg\min_{\boldsymbol{\beta}} \mathcal{L}_{\mathrm{CE}}(r, \hat{r}) + \mu\|\boldsymbol{\beta}\|_1$$

Non-zero entries $\{k : \beta_{e,k} \neq 0\}$ identify the SAE features predictive of routing to expert e. Their descriptions $\{s_k\}$ are composed into a natural-language explanation via a language model.

Learning or Inference Procedure

1. Train a SAE on the frozen MoE's residual stream to obtain a dictionary of monosemantic features.

2. Collect routing labels $r_{t,e}$ over a large corpus.
3. Fit sparse probes $\boldsymbol{\beta}_e$ for each expert over SAE feature activations.
4. Extract the top-k predictive features per expert.
5. Pass the feature descriptions to a language model to compose a coherent expert summary.

What the Guarantee Says

RouterInterp provides an empirical claim validated by causal intervention: activating a high-weight SAE feature at a token that would otherwise not route to expert e measurably increases routing probability to e, confirming causality beyond mere correlation.

Experimental Findings

- On **gpt-oss-20b**, RouterInterp achieves **≈65% higher routing-detection accuracy** than token-statistics baselines.
- Manual inspection confirms that expert descriptions are specific, coherent, and capture non-obvious expert roles.
- The SSH is validated: the top SAE features per expert span semantically disjoint categories (e.g., medical terminology *and* certain syntactic patterns in Spanish for the same expert).
- Results are consistent across all expert layers in gpt-oss-20b.

Ablations and Interpretation

Raw activations vs. SAE features: Probing on raw residual stream activations degrades routing prediction accuracy significantly and yields incoherent explanations.

Top-k features: Including more features beyond the natural coefficient elbow adds noise without improving routing accuracy.

Causal intervention: Artificially activating a predictive SAE feature causally increases routing probability to the predicted expert.

Across layers: Early layers show routing correlated with syntactic SAE features; later layers show richer semantic specialisation.

Reference

Ilya Lasy, Nora Yinuo Cai, Kola Ayonrinde. **RouterInterp: Understanding Superposed Specialisation in Mixture of Experts Routing**. ICML 2026. arXiv:2610.11775, October 2026.
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